Sahaj Bhadja

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Summary

Software engineer experienced in **Java, Linux, Git, CI/CD, and distributed systems**, writing tested APIs and cloud services with Kubernetes, Docker, and TCP/IP fundamentals, plus code reviews and design documentation.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Built fault-tolerant async job orchestration with FastAPI workers, processing 1,000+ jobs with explicit state transitions and failure isolation, increasing concurrent throughput 3x under distributed load at scale.
- Implemented cloud data pipelines processing 5,000+ monthly jobs with validation, reconciliation, and CI/CD-backed releases, achieving 99%+ reliability across testing and rollout phases at production scale end to end.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Wrote tested Python and JavaScript services serving 150+ instructors and processing 10,000+ daily requests, participating in code reviews while maintaining stable API performance end to end at production scale.
- Implemented automated tests and documented fallback paths for OCR and API failures, reducing customer feedback turnaround time 60% during integration testing and rollout reliably at production scale.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Coded object-oriented Java microservices with 20+ unit and integration tests on Linux, Windows, and macOS, using Jenkins CI/CD and Git while increasing pipeline efficiency 68% at scale.
- Participated in Agile code reviews and design discussions for API integrations, improving response times 30% while reducing technical debt 20% across production Linux workflows end to end.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed REST APIs and eight Docker-packaged microservices, syncing two million records while reducing resource wastage 42% and increasing conversions 15% for operational users reliably at production scale.
- Participated in code reviews and Jenkins CI/CD rollouts, cutting deployment time 50% while using Sentry to debug production issues across cloud-hosted services end to end at scale.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Contributed design documentation and architecture recommendations after mapping 12+ integration points, then implemented Flask APIs for booking and payment workflows serving end users reliably at production scale.
- Delivered tested Flask and Firebase services across 50,000+ records, validating implementation against specifications while achieving sub-200-millisecond responses and reducing manual refund handling 70% for customers at production scale.

Skills

Languages: Java, C++, Python, JavaScript, SQL
Systems: Linux, Git, Bash, Docker, Kubernetes, Maven
Cloud: AWS, Azure, GCP, CI/CD, Jenkins, GitHub Actions
Networking: TCP/IP, REST APIs, JSON, XML, YAML
Backend: FastAPI, Flask, Node.js, Spring Boot, PostgreSQL, MySQL, Distributed Systems
Practices: Code Review, Unit Testing, Integration Testing, Agile, Scrum, JIRA, Mockito

Projects

Chat2SQL Code | *Java, Linux, Open Source*
- Built Java and TypeScript services on Linux, Windows, and macOS with Docker and Git, configuring schema validation and reliable cross-platform integration for 10+ databases at production scale.

FinNexus Live | *Kubernetes, Distributed Services*
- Implemented Kubernetes deployment automation and CI/CD pipelines for a TypeScript platform supporting 50,000+ users and one million transactions, maintaining reliable production delivery with MongoDB persistence at scale.

CPU-scheduling Simulator Live | *C++, Computer Architecture*
- Implemented object-oriented C++ scheduling algorithms including FCFS, SJF, SRT, and Round Robin with React controls, reducing average turnaround time 25% across 1,000+ simulated process combinations at scale.